

PHYS 230 - Electricity and Magnetism

Lecture 8

Instructor: Saeed Rastgoo



**UNIVERSITY
OF ALBERTA**

Outline

2 Gauss's Law

- Electric Flux

All figures in these slides are from one of the following sources, unless cited otherwise:

University Physics With Modern Physics, Young & Freedman

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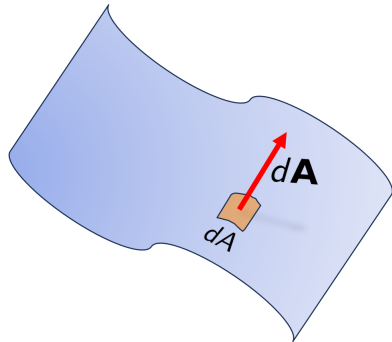
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Infinitesimal Area Vector

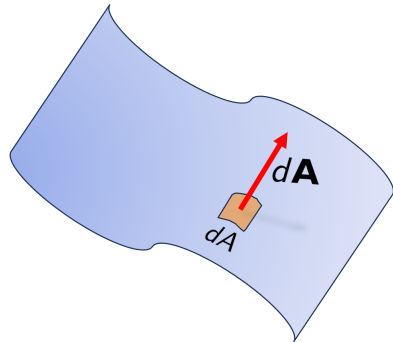
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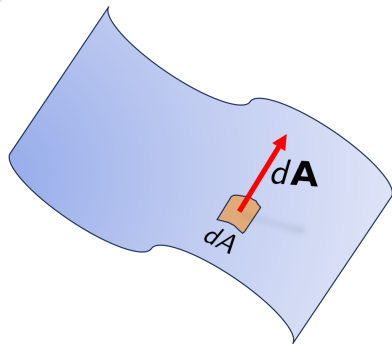
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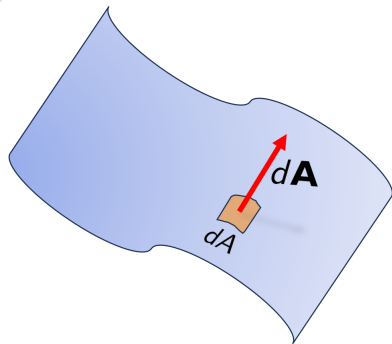


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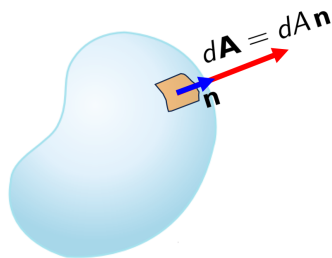
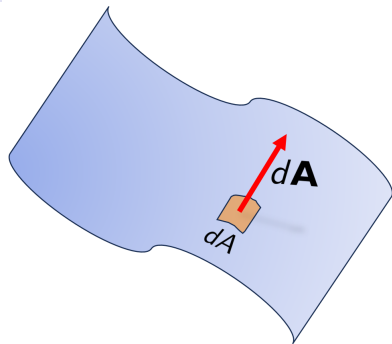
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- For a **closed surface**, i.e., a **surface enclosing a volume** $\Rightarrow$ $\mathbf{n}$ and thus $d\mathbf{A}$ are **pointing outwards** the volume



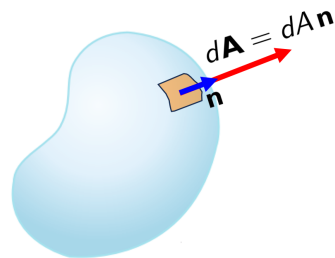
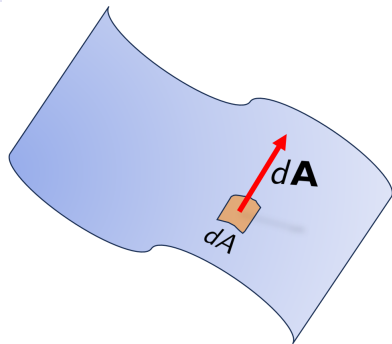
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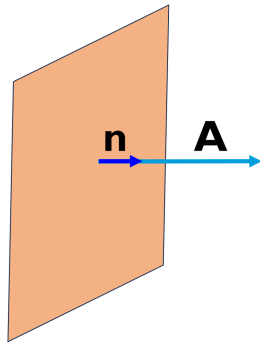
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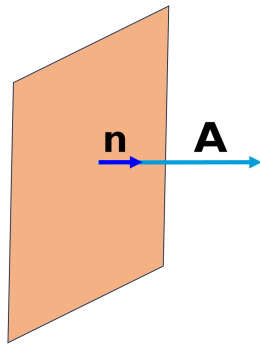
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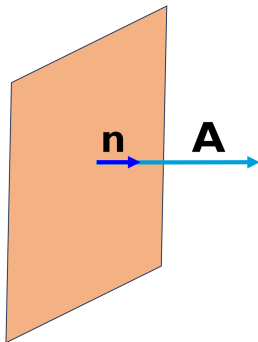
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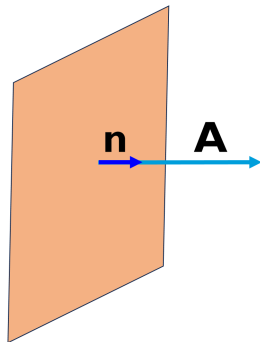


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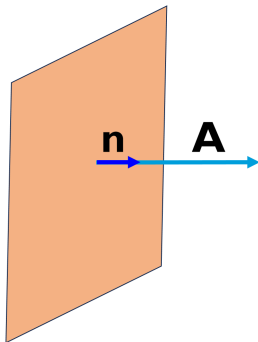
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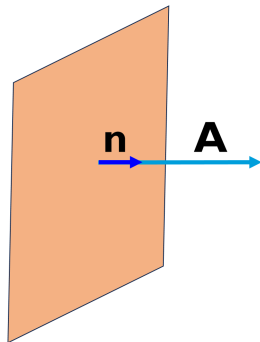
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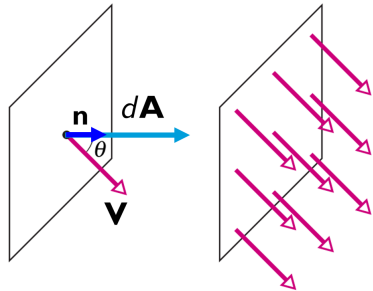
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Flux of a Vector Field

Given an area $d\mathbf{A}$ and a vector field $\mathbf{V}$, the infinitesimal flux $d\Phi$ of $\mathbf{V}$ through $d\mathbf{A}$ is

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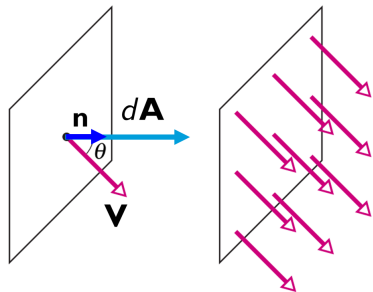


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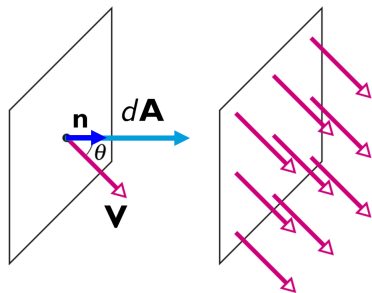


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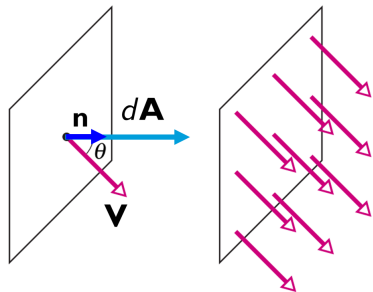


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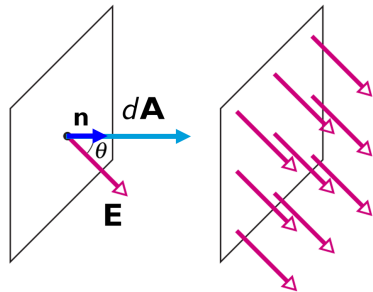


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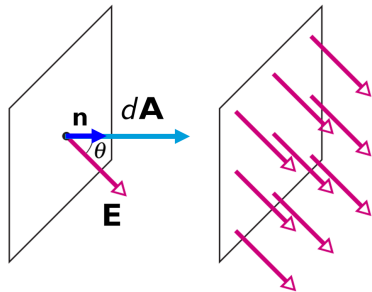
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- The units of flux if $\mathbf{V} = \mathbf{E}$ is

$$[d\Phi] = [\mathbf{E}] [d\mathbf{A}] = \frac{\text{Nm}^2}{\text{C}}$$

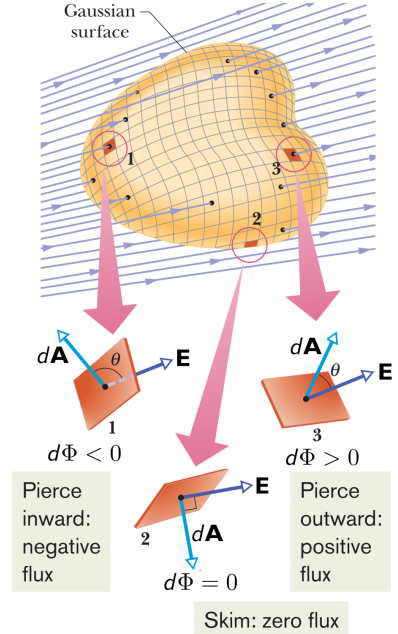


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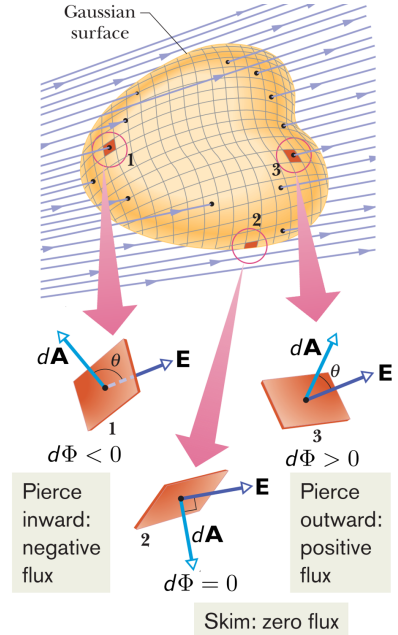


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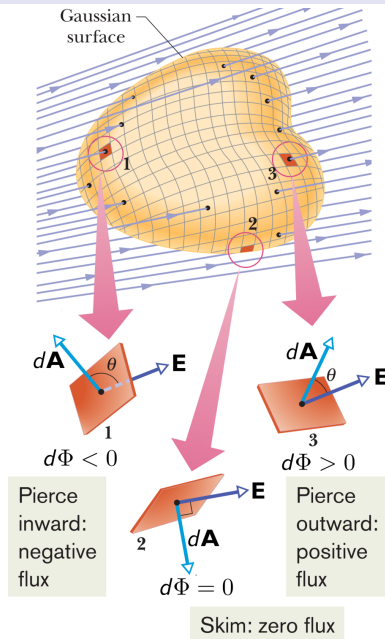


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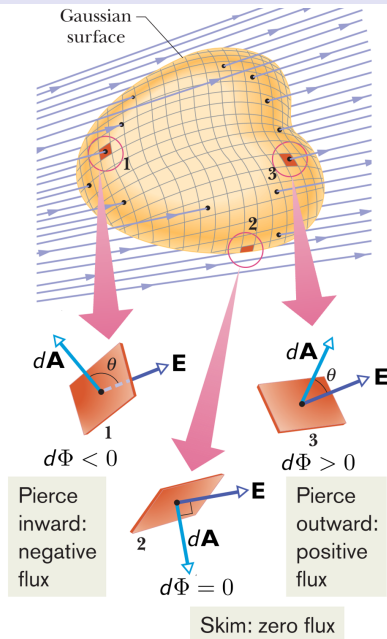


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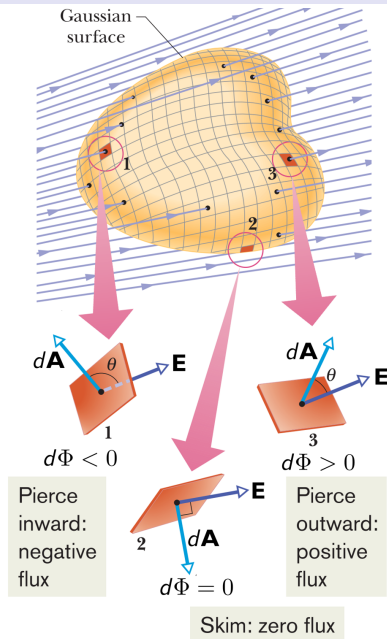


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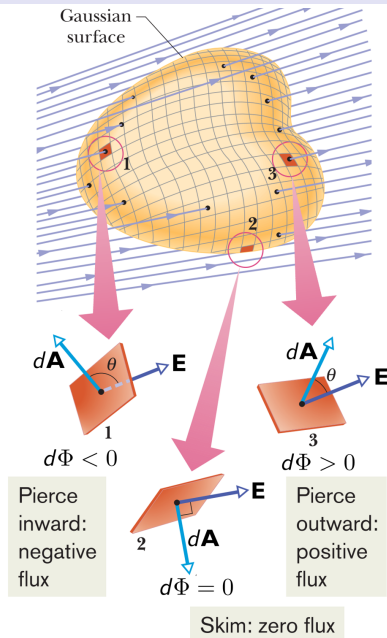
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- If $\mathbf{V}$ is **NOT constant** over the surface, the total flux of $\mathbf{V}$ passing through the whole area $\mathbf{A} = \int d\mathbf{A}$ is

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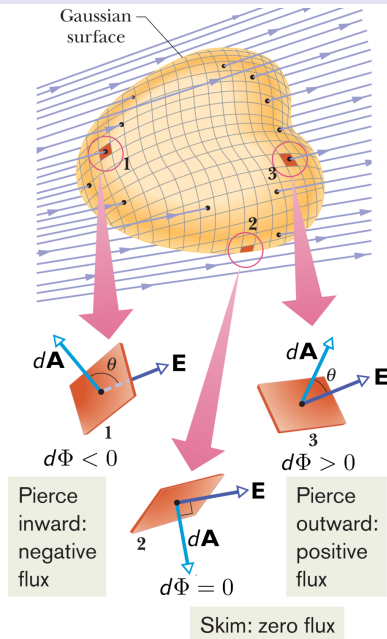
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- If $\mathbf{V}$ IS constant over the surface, the total flux of $\mathbf{V}$ passing through the whole area $\mathbf{A} = \int d\mathbf{A}$ is

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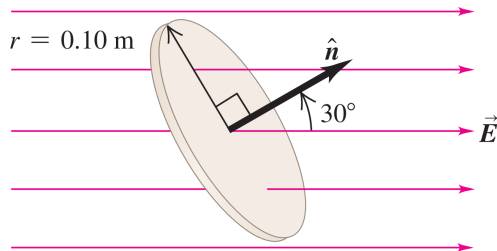


Exercise I

<https://epoll.srv.ualberta.ca> – Code: MBD

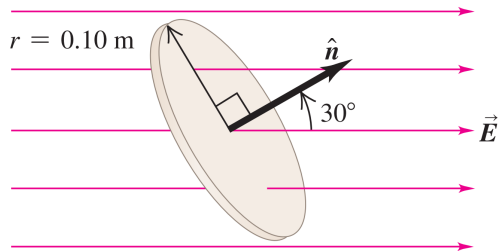
A disk of radius 0.10 m is oriented with its normal unit vector $\mathbf{n}$ at 30° to a uniform electric field $\mathbf{E}$ of magnitude $2 \times 10^3\text{ N/C}$ as in figure. **Since this is not a closed surface, it has no “inside” or “outside”! That is why we have to specify the direction of $\mathbf{n}$ in the figure.**

- (a) What is the electric flux through the disk?
- (b) What is the flux through the disk if it is turned so that $\mathbf{n}$ is perpendicular to $\mathbf{E}$?
- (c) What is the flux through the disk if $\mathbf{n}$ is parallel to $\mathbf{E}$?



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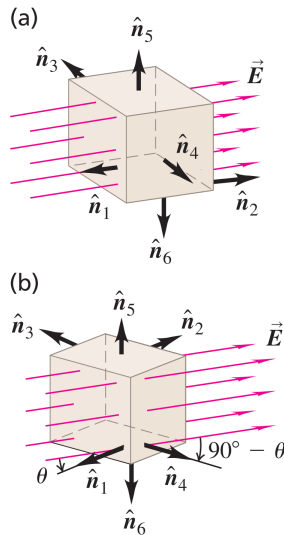
Exercise 2

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An imaginary cubical surface of side L is in a region of uniform electric field $\vec{E}$. Find the electric flux through each face of the cube and the total flux through the cube when

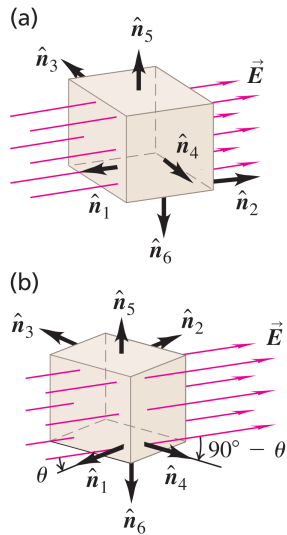
(a) the cube is oriented with two of its faces perpendicular to $\vec{E}$ (figure a)

(b) the cube is turned by an angle θ about a vertical axis (figure b)



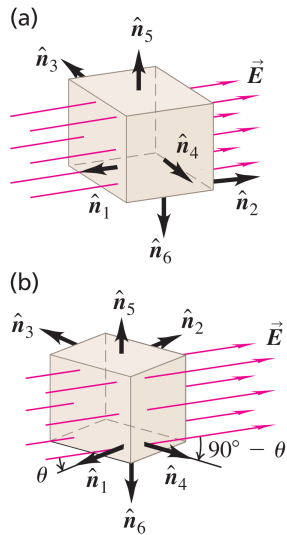
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